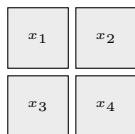


1. Patchify the image
into a sequence



patch-token sequence

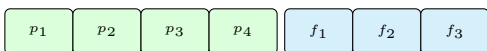


2. Encode patch and feature blocks
in one optical state

Patch modes and feature modes
share one 2-photon interferometer

patch modes

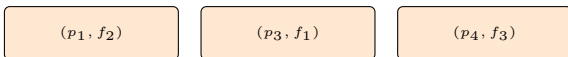
feature modes



two photons interfere across both mode blocks

3. Read out sectors to form
updated patch features

cross sector $\Omega_{\times}$ gives patch-feature interactions



same-block sectors add extra correlations



D: keep only
cross outputs

D_full: keep cross +
same-block outputs

Why this is transformer-like. The image is still patchified into a sequence, and the block still returns updated patch features. Classical ViT attention forms an explicit attention matrix; the compound models instead let multi-photon interference decide which patch-feature interactions survive the readout.